

Nevò MIRZAI HAMADANI

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PROFESSIONAL SUMMARY

Currently interning at Embodied AI, a startup focused on the development of humanoid robots, where my work centers on the teleoperation stack. I am pursuing a Master's in Robotics at EPFL with a minor in Data Science, where I have served as Teaching Assistant across multiple courses and led the simulation team in the EPFL Rocket Team, focusing on vertical landing simulation. My interests lie in mobile robotics and learning algorithms.

EDUCATION

EPFL | Master in Robotics (2nd year) Sep. 2024 – July 2026

- Minor in Data Science: Reinf. Learn.: 6/6, Deep Learn.: 5.75/6, Image analysis: 5.75/6
- Relevant courses: Robotic Manip.: 5.5/6, Aerial Robotics: 5.5/6, Legged Robots: 5.25/6

EPFL | Bachelor in Mechanical Engineering Sep. 2021 – July 2024

- 1 year exchange at the Chinese University of Hong Kong (CUHK)
- Relevant courses: Linear Algebra: 5.75/6, Analysis: 5.25/6, Algorithms: 5.75/6

TECHNICAL SKILLS

 **Programming** Python, C++, C, MATLAB  **Technical** ROS2, Docker, CUDA, Git

 **Simulation** IsaacSim, Mujoco, Gazebo, Simulink, Gymnasium, Unity, Genesis, Webots

 **Machine Learning** Reinforcement Learning, Deep Learning, Pytorch, IsaacLab

 **Computer Vision** SLAM (Visual/LiDAR), Image Analysis, OpenCV

 **Laboratory** Soldering, electrical circuits, CAD, 3D Printing

EXPERIENCE

Embodied AI | Robotics Internship, Humanoids Feb. 2026 – Aug. 2026

- Developed a ROS2/Docker-based teleoperation pipeline for an integrated exosuit. Currently scaling data acquisition to train Imitation Learning models for autonomous domestic tasks (vacuuming, household chores).

EPFL | Teaching Assistant, Student Mentor Sep. 2022 – Dec. 2025

- *Teaching Assistant:* Exercise sessions for courses such as Algorithms, Programming, Mechatronics, Linear Algebra, Calculus, and Materials.
- *Student Mentor:* Mentored freshmen throughout the academic year.

EPFL Rocket Team | Project Lead: Flight Simulation Sep. 2021 – June 2024

Vertical landing rocket project (Icarus) - Simulink, MATLAB

- Led the flight simulation team in developing the flight dynamics and vertical landing models using Simulink and MATLAB. Currently transitioning to real flight testing.

RESEARCH PROJECTS

Hand Gesture Drone Swarms Control - LIS Lab EPFL

- Conceived an intuitive system for controlling a swarm of drones using hand gestures.
- Controlled a simulated swarm in Unity, used ROS2 to transfer control to a real swarm.

3D Printing Drone Simulation - EMPA LSR Lab - Waiting for Paper Publication

- Constructed a simulation framework for a 3D printing drone using the Genesis engine.
- Modeled viscoelastic curing and expanding fluids using the Material Point Method by modifying the physics simulation engine at its core.

CORE AI PROJECTS

Imitation Learning with Robotic Arm

- Assembled a leader-follower robotic arm system.
- Implemented an ACT (Action Chunking with Transformers) policy.
- Designed a vision-based control pipeline using ResNet-18 encoders.

Reinforcement Learning for Quadruped Robot - RL, OpenAI Gym, Isaac Gym

- Trained a SAC Reinforcement Learning policy to achieve quadruped robot locomotion.
- Achieved high/low velocity gait control and stair climbing.

Inverse RL deep research - Reinforcement Learning

- Conducted research into the limitations and possible solutions of current IRL methods.

Data Creation Box - Object Recognition, System Integration

- Developed an automated physical environment to generate diversified organic datasets. Fine-tuned object recognition models and deployed them on a robotic arm system.

AI model training for Hate Speech prediction - GPT2/BERT Fine-tuning

- Fine-tuned GPT-2 and BERT to predict the probability that hate speech will be used in a phrase given the context, then compared the results.

Chocolate recognition - Neural Network training from scratch

- Selected and trained from scratch a light-weight neural network to recognize and count the number of different chocolates in multiple images with varying backgrounds.

CORE ROBOTIC PROJECTS

Robotic Arm Simulation for Scaffolding Applications in Hong Kong - Isaac Sim

- Simulated inverse kinematics for a robotic arm to assist in scaffolding construction.

Autonomous drone racing - Path planning, Hardware

- Programmed a drone to race a given trajectory autonomously, first in simulation and then deployed on a real Crazyflie (Sim2Real). Path planning, Computer Vision.

Autonomous Vehicle Navigation System Using MPC - Control, Simulink

- Devised a simplified autonomous vehicle navigation system in simulation (MPC).

Eye Tracking Headlamp - Robotic Manipulation, Computer Vision

- Prototyped a moving headlamp using an IR camera system for real-time eye tracking.
- Engineered a closed-loop system to automatically align lighting with the user's eyes.

Obstacles Dodging and Navigation for a Robot - State Estimation, CV, Path plan.

- Implemented vision (OpenCV), Kalman filter, and path planning for a Thymio robot.

Quadcopter Design and Stabilization - Robotic Manipulation, Control, Design

- Programmed PID stabilization from scratch using an MPU and an Arduino.

AWARDS - INTERESTS - LANGUAGES

★ Guinness World Record (2017): Most drones built and flown simultaneously (U18)

🏆 Talent Kick (2026): Selective cross-university high-tech startup incubator.

🗣️ English C1, French C1, Italian C2, German B1 🍻 Judo (competed internationally)